Q1 ALY 3510 · Question

Which project is credited with formally launching artificial intelligence as an organized research field?

A. The publication of Turing's "Computing Machinery and Intelligence" (1950)

B. The founding of the MIT AI Laboratory

C. The first International Joint Conference on AI

D. The Dartmouth summer research project (1956)

Q2 ALY 3510 · Question

What was the Turing Test originally designed to evaluate?

A. Whether a machine is conscious and self-aware

B. Whether a machine can calculate faster than a human

C. Whether a machine's conversational behavior is indistinguishable from a human's

D. Whether a machine can learn without being programmed

Q3 ALY 3510 · Question

Which statements accurately describe the "AI Winter" periods? (Select ALL that apply)

A. Ambitious early predictions about AI went largely unfulfilled

B. Researchers concluded AI was impossible and permanently abandoned the field

C. Government and private funding for AI research declined

D. Hardware limitations restricted what AI systems could accomplish

Q4 ALY 3510 · Question

Which statements correctly distinguish the major approaches to AI? (Select ALL that apply)

A. Symbolic AI learns patterns automatically from large datasets

B. Machine learning derives patterns from data rather than relying only on hand-written rules

C. Neural networks are loosely inspired by networks of neurons in the brain

D. Symbolic AI represents knowledge through explicit symbols and rules

Q5 ALY 3510 · Question

A system stores the facts "Canary is-a Bird" and "Bird has Wings" as connected nodes and links, then infers that a canary has wings. Which knowledge representation is it using?

A. A frame with slots and default values

B. A neural network with learned weights

C. A relational database of definitions

D. A semantic network

Q6 ALY 3510 · Question

A "Restaurant" representation includes slots for cuisine, price range, and seating, with seating defaulting to "tables" unless stated otherwise. What makes this structure useful?

A. It eliminates the need for any reasoning about the restaurant

B. It allows the computer to generate new restaurants automatically

C. It organizes related knowledge into a structured package that can supply expected values

D. It stores knowledge as weights learned from examples

Q2 ALY 3510 · Answer

Answer: C

Concept: AI history

You chat with someone you can't see. If you can't tell whether it's a person or a computer, the computer passes. The test only checks behavior. It says nothing about whether the machine is aware of anything.

Q1 ALY 3510 · Answer

Answer: D

Concept: AI history

In 1956, researchers met at Dartmouth for a summer and gave the field its name, "artificial intelligence." That meeting is the field's birthday. Turing's 1950 paper came first, but it was one person's idea, not the launch of a field.

Q4 ALY 3510 · Answer

Answers: B, C, D

Concept: AI paradigms

Symbolic AI follows rules people write down. Machine learning figures out the rules from examples. Neural networks are machine learning loosely modeled on brain cells. A is wrong because it describes machine learning, not symbolic AI.

Q3 ALY 3510 · Answer

Answers: A, C, D

Concept: AI history

People promised too much, the computers were too weak to deliver, and the money dried up. That's an AI Winter. B is wrong: research slowed down but never stopped for good. "Permanently" is the giveaway.

Q6 ALY 3510 · Answer

Answer: C

Concept: Knowledge representation

A frame is a form with blanks to fill in. If nobody fills in "seating," it assumes tables, because that's what you'd usually expect. Think fill-in-the-blank form with default answers.

Q5 ALY 3510 · Answer

Answer: D

Concept: Knowledge representation

Picture dots connected by lines: Canary to Bird to Wings. Follow the lines and you learn that a canary has wings. Dots and lines is a semantic network. Think graph.

Q7 ALY 3510 · Question

A clinician sees fever, cough, and fatigue and concludes that influenza is the most likely explanation. Which type of reasoning is this?

A. Abductive

B. Deductive

C. Inductive

D. Analogical

Q8 ALY 3510 · Question

Which statements about reasoning types are accurate? (Select ALL that apply)

A. Deductive reasoning produces conclusions that must be true if the premises are true

B. Inductive reasoning always produces certain conclusions

C. Analogical reasoning infers similarity in one respect from similarity in others

D. Predictive reasoning estimates outcomes that have not yet occurred

Q9 ALY 3510 · Question

A model is trained on thousands of emails already labeled "spam" or "not spam." What kind of learning is this?

A. Supervised learning

B. Unsupervised learning

C. Reinforcement learning

D. Symbolic reasoning

Q10 ALY 3510 · Question

A model scores very well on its training data but poorly on new, unseen data. What is the most likely problem?

A. Underfitting

B. Too much validation data

C. Successful generalization

D. Overfitting

Q11 ALY 3510 · Question

Which scenarios correctly match a type of machine learning? (Select ALL that apply)

A. Grouping customers into segments with no predefined labels: unsupervised learning

B. Predicting tomorrow's temperature as a number: classification

C. A game-playing agent improving through rewards and penalties: reinforcement learning

D. Predicting whether a loan applicant will default (yes or no): classification

Q12 ALY 3510 · Question

Why is a separate test dataset held back until after a model is trained?

A. To give the model additional training examples

B. To estimate how well the model generalizes to data it has never seen

C. To guarantee the model will perform perfectly in the real world

D. To remove the need for validation data

Q8 ALY 3510 · Answer

Answers: A, C, D

Concept: Reasoning

Deductive means true rules give a guaranteed answer. Analogical means "this is like that." Predictive means guessing what happens next. B is wrong: inductive reasoning generalizes from examples, so it's probable, never certain. "Always" is the giveaway.

Q7 ALY 3510 · Answer

Answer: A

Concept: Reasoning

Abduction is the detective's guess: "What best explains what I'm seeing?" The doctor isn't certain, but flu explains all three symptoms best. Best explanation means abductive.

Q10 ALY 3510 · Answer

Answer: D

Concept: Machine learning

It memorized the practice test instead of learning the subject. It aces questions it has seen and fails new ones. That's overfitting. Underfitting would mean it does badly on both.

Q9 ALY 3510 · Answer

Answer: A

Concept: Machine learning

Every email comes with the answer attached, like flash cards with the answer on the back. Learning from labeled examples is supervised learning.

Q12 ALY 3510 · Answer

Answer: B

Concept: Machine learning

You keep some questions secret so the real exam is fair. The test set shows how the model does on things it has never seen. It estimates performance; it can't guarantee it, which rules out C.

Q11 ALY 3510 · Answer

Answers: A, C, D

Concept: Machine learning

Sorting without labels is unsupervised. Learning from points and penalties is reinforcement. Yes-or-no is classification. B is wrong: predicting a number like a temperature is regression, not classification.

Q13 ALY 3510 · Question

Which statements about how machines and humans learn are accurate? (Select ALL that apply)

A. Humans can often grasp a new concept from only a few examples

B. Many ML systems require large amounts of data to learn reliably

C. Machine learning always needs less data than humans to learn a concept

D. Humans can never learn from a single example

Q14 ALY 3510 · Question

During neural network training, what happens after the network's output is compared with the correct answer?

A. The weights are adjusted to reduce the error

B. The network permanently stops learning

C. New input data is generated automatically

D. The number of layers is reduced

Q15 ALY 3510 · Question

Why do neural networks use nonlinear activation functions?

A. To make every neuron produce the same output

B. To eliminate the need for training data

C. To let the network model complex, nonlinear relationships

D. To make the network's decisions fully transparent

Q16 ALY 3510 · Question

Which statements describe what deep networks do across their layers? (Select ALL that apply)

A. They require programmers to define each feature by hand

B. Early layers tend to capture simple features while later layers combine them into more abstract ones

C. They learn hierarchical representations directly from data

D. Each layer's output becomes input to the next layer

Q17 ALY 3510 · Question

What is the key difference between a neural network's learned weights and a symbolic system's rules?

A. Weights are written by programmers; rules are learned

B. Weights and rules are identical in function

C. Rules can never be updated once written

D. Weights are adjusted from data during training; rules are explicitly specified by humans

Q18 ALY 3510 · Question

Which concerns arise when deep learning systems are used in high-stakes decisions? (Select ALL that apply)

A. It can be hard to explain why the system reached a particular decision

B. Deep learning systems are always less accurate than human experts

C. Biases in training data can be reproduced in the system's outputs

D. Opacity is a concern only for systems that never make mistakes

Q14 ALY 3510 · Answer

Answer: A

Concept: Neural networks

Guess, check, adjust, repeat. The network measures how wrong it was, then tweaks its weights a little so the next guess is closer. That's how it learns.

Q13 ALY 3510 · Answer

Answers: A, B

Concept: Human vs. machine learning

A kid can learn "giraffe" from one or two pictures. Many AI systems need thousands. That gap is called sample efficiency. C and D use "always" and "never," and both are false.

Q16 ALY 3510 · Answer

Answers: B, C, D

Concept: Deep learning

Each layer passes its work to the next. Early layers find simple things like edges. Later layers combine those into shapes, then objects. The network learns these features itself, so A is wrong.

Q15 ALY 3510 · Answer

Answer: C

Concept: Neural networks

Without them, a network can only draw straight lines, no matter how many layers it has. Activation functions let it draw curves, so it can learn messy, real-world patterns.

Q18 ALY 3510 · Answer

Answers: A, C

Concept: Ethics

It's a black box: it can be hard to say why it decided something. And if the examples it learned from were unfair, its answers can be unfair too. B uses "always" and is false. D doesn't make sense.

Q17 ALY 3510 · Answer

Answer: D

Concept: Neural networks vs. symbolic AI

A person writes the rules. The computer learns the weights. A flips this around, which makes it a trap. C uses "never," and people can rewrite rules anytime.

Q19 ALY 3510 · Question

What do distributed AI systems spread across multiple agents, devices, or organizations?

A. Intelligence or computation that would otherwise sit in one central system

B. Only data storage, while all decisions stay with one controller

C. Identical copies of a single agent that never interact

D. Only user interfaces, while computation stays centralized

Q20 ALY 3510 · Question

Which statements describe advantages commonly associated with distributing AI? (Select ALL that apply)

A. A distributed system is completely immune to failure

B. Work can be processed in parallel across many nodes

C. The system can continue operating if an individual node fails

D. Computation can happen closer to where data is generated

Q21 ALY 3510 · Question

When an outcome emerges from many interacting agents owned by different organizations, which governance challenges become more significant? (Select ALL that apply)

A. Determining who is responsible when something goes wrong

B. Distribution always eliminates the need for regulation

C. Agreeing on who owns and may use shared data

D. Ensuring agents from different makers can communicate through shared standards

Q22 ALY 3510 · Question

Which statement best describes the role of AI frameworks such as TensorFlow or PyTorch?

A. They replace the need for data in model development

B. They provide reusable building blocks that make developing and training models faster and more accessible

C. They guarantee that every model built with them is unbiased

D. They are used only by expert researchers and never by non-specialists

Q23 ALY 3510 · Question

A facilities engineer uses an AI platform to build a predictive maintenance model with help from a data scientist. Which statements reflect how frameworks shape this collaboration? (Select ALL that apply)

A. The engineer can focus on defining the problem and which equipment data matters

B. The data scientist's technical role becomes completely unnecessary

C. Shared tools make it easier for the two disciplines to work together

D. The engineer's domain knowledge helps judge whether the model's results make sense

Q24 ALY 3510 · Question

A classifier labels the review "This is not good at all" as positive because it contains the word "good." What does this suggest about the classifier?

A. It understands context better than a neural model

B. It is correctly detecting sarcasm

C. It has access to the reviewer's intentions

D. It treats words independently and ignores word order and negation

Q20 ALY 3510 · Answer

Answers: B, C, D

Concept: Distributed AI

Many hands make light work. They can work at the same time, keep going if one stops, and do the job right where the data is. A is wrong: "completely immune" is too absolute.

Q19 ALY 3510 · Answer

Answer: A

Concept: Distributed AI

Instead of one big brain doing all the thinking, the thinking is shared among many smaller helpers. The answers that start with "only" are traps.

Q22 ALY 3510 · Answer

Answer: B

Concept: AI frameworks

They're like LEGO kits. You get ready-made pieces, so you don't build every brick from scratch. You still need data, and the kit can't guarantee fairness.

Q21 ALY 3510 · Answer

Answers: A, C, D

Concept: Distributed AI governance

When many companies' systems work together, it's harder to say who's to blame, who owns the data, and whether the systems speak the same language. B uses "always" and is false.

Q24 ALY 3510 · Answer

Answer: D

Concept: NLP

It looks at each word by itself, like counting words in a bag. It saw "good" and missed the "not" in front of it. That's how a bag-of-words model like Naive Bayes works.

Q23 ALY 3510 · Answer

Answers: A, C, D

Concept: AI frameworks

The tools let the engineer focus on what matters, like which readings point to a failing pump, and help the two people work together. B is wrong: the expert still matters. "Completely" is the giveaway.

Q25 ALY 3510 · Question

Which statements about how large language models handle language are accurate? (Select ALL that apply)

A. They learn statistical patterns from very large amounts of text

B. They store a hand-written rule for every grammatical construction

C. Their fluency does not by itself settle whether they understand meaning

D. They can produce coherent text without an explicitly programmed knowledge base

Q26 ALY 3510 · Question

Which challenges should a company consider before deploying a chatbot worldwide? (Select ALL that apply)

A. Every language follows the same grammatical rules, so one model fits all

B. Idioms and humor may not carry the same meaning across cultures

C. Lower-resource languages may get weaker performance than widely spoken ones

D. Translation can lose cultural nuance

Q27 ALY 3510 · Question

What does the success of LLMs suggest from the perspective of cognitive science and philosophy?

A. Human language understanding has been fully explained

B. Consciousness is required for any useful language output

C. Fluent language behavior may emerge from patterns without explicit knowledge representation, leaving open what counts as understanding

D. Language can never be modeled computationally

Q28 ALY 3510 · Question

Which ethical issues arise when AI can generate realistic text, voices, and images? (Select ALL that apply)

A. Uncertainty about who owns or deserves credit for AI-generated work

B. Generated content can always be reliably detected

C. The ease of producing persuasive false information at scale

D. People may not know whether they are communicating with a human or a machine

Q29 ALY 3510 · Question

As adaptive AI becomes common in workplaces and schools, which implications are reasonable to expect? (Select ALL that apply)

A. The skills and knowledge that education emphasizes may need to be rethought

B. Human judgment will become completely irrelevant

C. Questions arise about how much decision-making to delegate to AI

D. The nature of professional expertise may shift

Q30 ALY 3510 · Question

In which area does human intelligence generally hold an advantage over current AI?

A. Processing millions of records in seconds

B. Understanding emotions and responding with empathy

C. Performing repetitive calculations without fatigue

D. Applying the same rule identically every time

Q26 ALY 3510 · Answer

Answers: B, C, D

Concept: NLP and culture

Jokes and sayings don't always translate. Languages with less online text get worse tools. Meaning can get lost in translation. A is wrong: languages have different grammar. "Every" is the giveaway.

Q25 ALY 3510 · Answer

Answers: A, C, D

Concept: LLMs

They learn by reading a huge amount of text and picking up the patterns. They write well, but writing well doesn't prove they understand. Nobody hand-wrote their grammar rules, so B is wrong.

Q28 ALY 3510 · Answer

Answers: A, C, D

Concept: Ethics

Who gets credit? Fake news gets cheap and easy to make. You might not know you're talking to a bot. B is wrong because AI content often can't be detected reliably. "Always" is the giveaway.

Q27 ALY 3510 · Answer

Answer: C

Concept: LLMs and understanding

LLMs show that good writing can come from patterns alone, with nobody programming in the facts. Whether that counts as understanding is still an open question. The other answers say "fully," "required," or "never."

Q30 ALY 3510 · Answer

Answer: B

Concept: Human vs. AI strengths

Computers win at speed, endurance, and consistency. People are better at understanding how others feel and caring about it.

Q29 ALY 3510 · Answer

Answers: A, C, D

Concept: Society and work

Schools may teach different skills, jobs may change, and we have to decide how much to let AI decide. B is wrong: people's judgment still matters. "Completely" is the giveaway.